

Question	Answer	Mark	Mark scheme	Additional guidance
21	2.5	P1	for forming an equation using the volume, eg $20wh = 300$ or $wh = 15$ oe or for $300 \div 20 (= 15)$	Can use any variables for height and width
		P1	for forming an equation using the surface area, eg $20w + 20w + 20h + 20h + hw + hw = 370$ oe or $20w + 20w + 20h + 20h + 30 = 370$ oe	Use of 30 in this equation implies 1st P1
		P1	for eliminating one variable, eg $40h + \frac{600}{h} + 30 = 370$ or $40h^2 - 340h + 600 (= 0)$ or $2h^2 - 17h + 30 (= 0)$	
		P1	for process to solve quadratic equation, eg $(2h - 5)(h - 6) (= 0)$ or $\frac{17 \pm \sqrt{(-17)^2 - 4 \times 2 \times 30}}{2 \times 2}$	Award P4 for answer of 6 if 2.5 and 6 identified and supported by working that involves both volume and surface area
		A1	for 2.5 oe	Do not award A1 for 2.5 if it is not supported by working that involves both volume and surface area